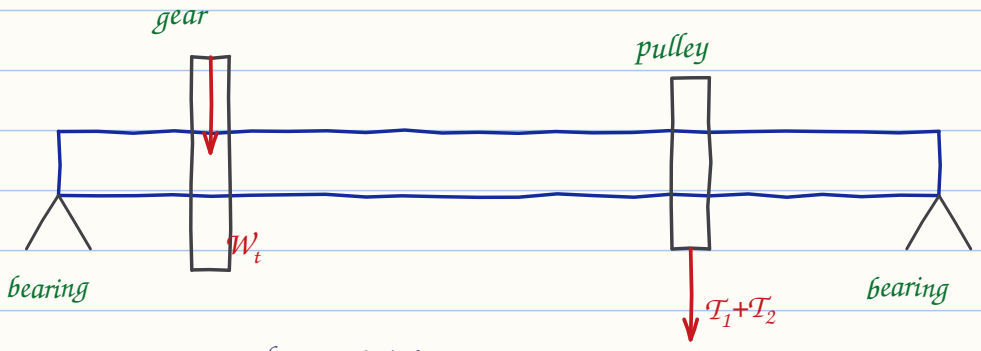
UNIT - III

Power Transmission Shafts and Couplings

1. Power Transmission Shafts

Shaft = rotating member (usually circular) that transmits power / torque and supports gears, pulleys, sprockets, flywheels. It is loaded by torsion, bending and sometimes axial load (helical gears, thrust).

- *Transmission shaft / line shaft* - carries power from the source to the machine ; *counter shaft* - intermediate shaft driven by a line shaft ; *spindle* - short shaft of a machine tool ; *axle* - supports rotating wheels, carries only bending (no torque).
- *Materials* : mild steel (C30, C40, 40C8, 45C8) - common ; alloy steel 40Cr1, 40Ni14, 35Mn2Mo28, 15Ni2Cr1Mo15 (case hardened) for high loads ; cold drawn / ground - better surface & accuracy up to 50 mm ; hot rolled - large sizes.
- *Requirements* : adequate strength (static & fatigue), rigidity (torsional & lateral), sizes from standard series : 6 8 10 12 14 16 18 20 22 25 28 30 32 36 40 45 50 55 60 65 70 75 80 90 100 mm...



$T = 9.55\times10^6\ P / N$ (N·mm)

M = max at gear / pulley section → shaft carries M and T together

Fig 1: A shaft with gear and pulley - bending + torque

Loads on the shaft

- *Torque* : $T = 60\ P / 2\pi N = 9.55\times10^6\ P(kW) / N(rpm)$ N·mm.
- *Gear* : tangential force $W_t = 2T/D$; radial $W_r = W_t \tan\varphi$ (spur) ; (helical : axial $W_a = W_t \tan\beta$ too).
- *Belt pulley* : $(T_1 - T_2)\ R = T$, bearing load = $T_1 + T_2$; $T_1/T_2 = e^{\mu\theta}$; for design $T_1 + T_2 \approx 1.5\text{-}2\ (T_1 - T_2)$ ... use given data.

- Resolve loads in vertical and horizontal planes, find reactions, draw B.M.D. in both planes, $M = \sqrt{(M_V^2 + M_H^2)}$ at the critical section ; draw T.M.D.

2. Design of Shafts - Strength Basis

(a) Shaft under torsion only

$$T = (\pi/16)\,\tau\, d^3 \quad \rightarrow \quad d^3 = 16\,T / \pi\,\tau_{allow}$$

hollow : $T = (\pi/16)\,\tau\, d_o^3\,(1 - K^4)$, $K = d_i/d_o$

- Allowable shear stress (ASME code) : $\tau_{allow} = 0.30\,S_y$ or $0.18\,S_{ut}$ (take the smaller). If a keyway is present reduce by 25% → $\tau_{allow} \times 0.75$.

Typical : 40C8 ($S_y = 330$, $S_{ut} = 580$) → $0.3 \times 330 = 99$; $0.18 \times 580 = 104$ → 99 MPa ; with keyway 74 MPa. For general shafting use τ = 40-60 MPa (machine-shop practice), small shaft with sudden shock : 20-30 MPa.

(b) Shaft under bending only (axle)

$$d^3 = 32\,M / \pi\,\sigma_{b,allow} \quad , \quad \sigma_{b,allow} = S_y/n \approx 0.6\,S_y / \ldots \text{(use given)}$$

(c) Combined bending and torsion - ASME code equation

Use maximum shear theory (ductile) with shock and fatigue factors K_b (bending) and K_t (torsion) which cover suddenly applied / reversing loads :

$$d^3 = 16 / (\pi\,\tau_{allow}) \cdot \sqrt{(\,(K_b\,M)^2 + (K_t\,T)^2\,)} \quad \text{(solid)}$$

$$d_o^3\,(1 - K^4) = 16 / (\pi\,\tau_{allow}) \cdot \sqrt{(\,(K_b\,M)^2 + (K_t\,T)^2\,)} \quad \text{(hollow)}$$

Type of load	K_b	K_t
Rotating shaft : gradually applied / steady	1.5	1.0
suddenly applied, minor shock	1.5 - 2.0	1.5 - 2.0
suddenly applied, heavy shock	2.0 - 3.0	1.5 - 3.0
Stationary shaft : gradually applied / steady	1.0	1.0
suddenly applied	1.5 - 2.0	1.5 - 2.0

Why : Rotating shaft → bending stress is completely reversed each revolution (fatigue !) so $K_b > 1$ even for steady load ; torque is steady → $K_t = 1$. That is the reason for $K_b = 1.5$, $K_t = 1.0$.

(d) With axial load F (tension / compression)

$$d^3 = 16 / (\pi \tau_{allow} (1 - K^4)) \cdot \sqrt{([K_b M + \alpha F d (1 + k^2)/8]^2 + (K_t T)^2)}$$

α = column factor = 1 for tension ; for compression of slender shaft $\alpha = 1/(1 - 0.0044 L/K)$ $(L/K < 115)$, K = radius of gyration. Axial load in short shafts is small - usually ignored. Equation is solved by trial (d appears on both sides) or neglect F first.

Example 1 : shaft transmits 30 kW at 360 rpm. Maximum bending moment 600 N·m (at gear). Material 40C8 (S_y = 330, S_{ut} = 580 MPa), keyway present. Rotating shaft, gradual load.

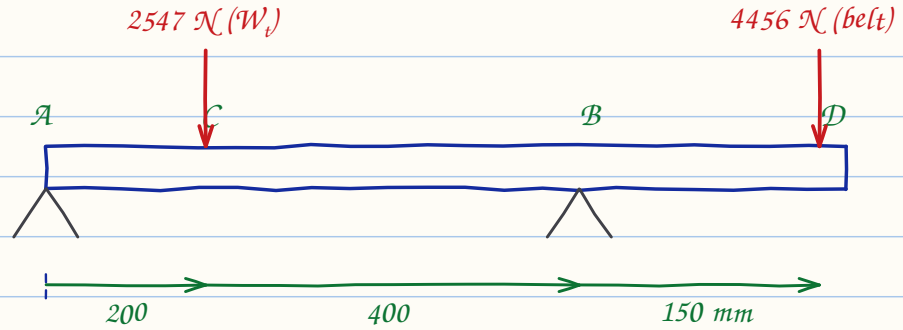
$T = 9.55\times10^6 \times 30/360 = 796\times10^3$ N·mm. τ_{allow} = min (0.3×330 , 0.18×580) = 99 → × 0.75 (keyway) = 74.3 MPa. K_b = 1.5, K_t = 1.

$d^3 = 16/(\pi \times 74.3) \sqrt{((1.5 \times 600\times10^3)^2 + (796\times10^3)^2)} = 0.0685 \times 1.2015\times10^6 = 82.4\times10^3$ → d = 43.5 mm → d = 45 mm (standard).

Hollow shaft (k = 0.5) : $d_o = d / (1 - K^4)^{1/3} = 43.5/0.979 = 44.4$ → provide d_o = 45 , d_i = 22.5 : weight ratio = $d_o^2 (1 - k^2) / d^2 = 0.78$ → 22% lighter for same strength. Hollow shaft has larger outer dia, hence higher cost / bearing size.

Worked example with gear and overhung pulley

Example 5 : 12 kW at 300 rpm. Gear (D = 300 mm, 20° pressure angle) at C, pulley (D = 400 mm, $T_1 = 2.5\ T_2$) overhung at D. Bearings A, B 600 mm apart. 40C8 with keyway (τ_{allow} = 74.3 MPa).



$T = 9.55\times10^6 \times 12/300 = 382\times10^3$ N·mm. Gear : $W_t = 2T/D = 2\times382\times10^3/300 = 2547$ N , $W_r = 2547 \tan 20° = 927$ N. Pulley : $T_1 - T_2 = 2T/D = 1910$ N → T_2 = 1273 , T_1 = 3183 → belt load $T_1 + T_2$ = 4456 N.

Take moments (V-plane) about A : $R_B \times 600 = 2547 \times 200 + 4456 \times 750$ → R_B = 6419 N , R_A = 584 N. B.M. at C = 584 × 200 = 117×10³ ; at B = 4456 × 150 = 668×10³ N·mm (overhang). H-plane : $M_C = 618\times200 = 124\times10^3$, M_B = 0. Resultant M_C =

170×10^3 N·mm , $M_B = 668\times10^3$ N·mm → section B is critical (torque also acts there).

Steady load ($K_b = 1.5$, $K_t = 1$) : $d^3 = 0.0685\sqrt{((1.5 \times 668\times10^3)^2 + (382\times10^3)^2)} = 73.5\times10^3$ → $d = 41.9$ mm → $d = 45$ mm.

Heavy shock ($K_b = 2.0$, $K_t = 1.5$) : $d^3 = 0.0685\sqrt{((2 \times 668\times10^3)^2 + (1.5 \times 382\times10^3)^2)} = 99.7\times10^3$ → $d = 46.4$ → $d = 50$ mm. Shock factors raised the size by 11%.

Example 5a (with axial load) : $M = 300$ N·m, $T = 200$ N·m, axial tension $F = 8$ kN, $\tau_{allow} = 60$ MPa, $K_b = 1.5$, $K_t = 1$, $\alpha = 1$, solid. Try $d = 35$: $F\,d/8 = 35\times10^3$ N·mm ; $K_b M = 450\times10^3$; $d^3 = (16/\pi 60)\sqrt{((450 + 35)^2 + 200^2)\times10^3} = 0.0849 \times 524.6\times10^3 = 44.5\times10^3$ → $d = 35.4$ mm → $d = 36$ mm (axial term raised d by < 2% - usually neglected for short shafts).

3. Shaft Design Based on Rigidity & Critical Speed

Torsional rigidity

$$\theta = T L / G J \text{ (rad)} \rightarrow \theta^\circ = 584\, T L / G d^4 \text{ (solid shaft)}$$
$$d^4 = 584\, T L / G \theta^\circ$$

- Allowable twist : 0.25°/m for machine tool shafts ; 0.5 - 1°/m for line shafts ; ≤ 1° per 20 d length for general transmission (precision drive).

Example 2 : same shaft ($T = 796$ N·m) , $L = 1$ m , $G = 80$ GPa , $\theta_{allow} = 0.5°/m$: $d^4 = 584 \times 796\times10^3 \times 1000 / (80\times10^3 \times 0.5) = 11.6\times10^6$ → $d = 58.4$ mm > 45 mm. So rigidity governs here → provide $d = 60$ mm. Always check both strength and rigidity.

Lateral deflection & critical speed

- Slope at bearings (≈ 0.001 rad for plain bearings ; 0.0005 for roller bearings) and deflection at gears (≤ 0.0005 L ..) maintain proper gear mesh and bearing life.
- Critical speed (whirling) : $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta)}$, $N_c = 946/\sqrt{\delta}$ rpm (δ = static deflection in mm under the loads). Operating speed should be < 0.8 N_c or > 1.3 N_c ; rigid shaft (below) vs flexible shaft (above N_c).

Fatigue design (fluctuating loads)

Rotating shaft : bending completely reversed ($\sigma_a = 32\, M K_f/\pi\, d^3$, $\sigma_m = 0$) + steady torque ($\tau_m = 16\, T/\pi\, d^3$). Use Unit I expression (Soderberg / von Mises) :

$$d^3 = (32\ n / \pi)\sqrt{\left((K_f M / S_e)^2 + (3/4)(T / S_y)^2 \right)}$$

Example 5b (fatigue) : rotating shaft, $M = 500$ N·m (reversed), $T = 400$ N·m (steady), $S_y = 380$, corrected $S_e = 200$ MPa, $K_f = 1.8$, $n = 2$. $d^3 = (32\times2/\pi)\sqrt{\left((1.8\times500\times10^3/200)^2 + 0.75\,(400\times10^3/380)^2 \right)} = 20.37 \times 4591 = 93.5\times10^3 \rightarrow d = 45.4$ mm $\rightarrow$ **$d = 50$ mm** (standard, next size up).

- Steps : (1) T from P and N ; (2) forces from gears / pulleys ; (3) V and H planes, B.M.D. ; (4) critical section ; (5) choose material, τ_{allow} (or S_e, K_f) ; (6) d from ASME or Soderberg ; (7) check torsional / lateral rigidity ; (8) standard size, keyway, shoulders with fillet $r \geq 0.1\ d$; (9) check critical speed.
- Design details : stepped shoulders 5-10% larger ; relief grooves ; key seat sled-runner ; interference fits ; use $K_f = 1.6$-2.0 for keyway in bending, 1.3-1.5 for shoulder fillet.

4. Keys, Keyways and Splines

A **key** is a machine element placed in a keyway between shaft and hub to prevent relative rotation and transmit torque. It is the weakest link - by design (cheap, sheared in overload).

Key type	Features
Sunk - rectangular / square	$w = d/4$, $t = 2w/3$ (rect) , $t = w$ (square) ; most common
Gib-head taper (1 in 100)	hub driven onto key, can be removed ; gives axial hold
Woodruff	half-disc, sits deep in shaft, self-aligning ; machine tools, tapers
Feather / sliding	fixed to shaft, hub slides axially ; gear shifting
Splines (involute / straight)	many integral keys, high torque, hub slides ; gear box

$$\text{Shear : } T = l\ w\ \tau_k\ (d/2) \rightarrow \tau_k = 2T / (d\ l\ w)$$

$$\text{Crushing : } T = l\ (t/2)\ \sigma_c\ (d/2) \rightarrow \sigma_c = 4T / (d\ l\ t)$$

$$\text{Equal strength with shaft (square key) : } l = (\pi/2)\ d\ \tau_s/\tau_k \approx 1.57\ d$$

- Key stock : 40C8 / 45C8 ($S_y \approx 300$; $\tau_k = 0.5\ S_y/n$). Key is weaker than the shaft in crushing $\rightarrow$ l is kept $\approx$ hub length (1.25 d - 1.5 d). For longer hubs use 2 keys at 120°-180° (each counted as 0.75).

- Keyway reduces shaft torsional strength by ~ 25% (ASME) and gives $K_f \approx 2.0$ (bending) → use sled-runner ends and radiused corners.

Example 6 : $d = 45$ mm, $T = 796$ N·m, key $14 \times 9 \times 60$: $\tau_k = 2\times796\times10^3/(45\times60\times14) = 42$ MPa (OK) ($\leq 0.5\, S_y/n$ for 40C8 key $= 165/2 = 82$) ; $\sigma_c = 4\times796\times10^3/(45\times60\times9) = 131$ MPa $\leq S_y/n = 165$ (OK).

5. Couplings

A coupling connects two shafts end to end to transmit torque while allowing for assembly, alignment error, shock absorption, and disconnection (clutch can be engaged during running, coupling only at rest).

Class	Types	Feature
Rigid	sleeve (muff), split muff, clamp (compression), flange (protected / unprotected)	shafts must be collinear
Flexible	bushed-pin, Oldham, chain, gear, diaphragm, rubber-tyre, grid	allows small misalignment, absorbs shock
Universal	Hooke's (cardan) joint, double Hooke's	intersecting shafts, angle ≤ 30°

- Requirements : easy assembly / disassembly, positive torque transfer (no slip), minimal weight (inertia), safe (no projecting bolt heads - use protective rim), able to absorb vibration, low maintenance.

Design torque : $T_d = K_s\, T$ where service factor K_s = 1.0-1.5 (electric motor drive) to 2.0-2.5 (engine / heavy shock). Mostly the rated T is multiplied by 1.5 for design.

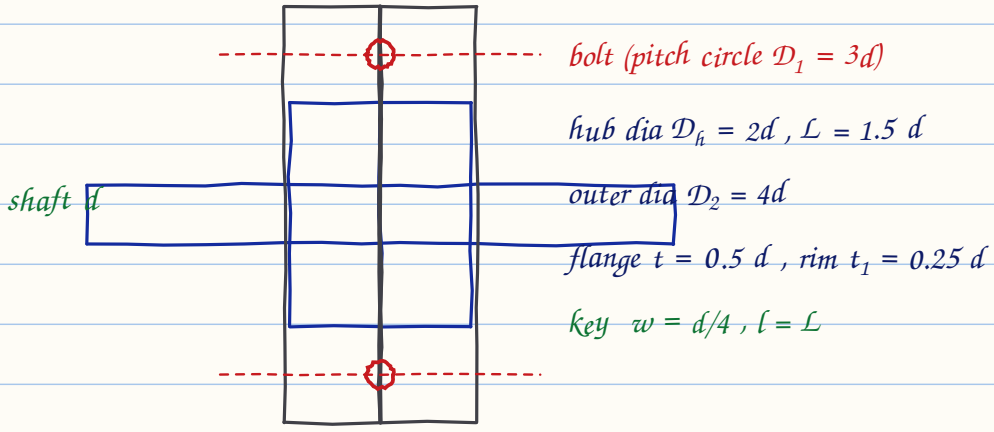
Comparison / selection of couplings

Coupling	Misalignment	Torque	Typical use
Muff / sleeve	none	low	light line shafts
Clamp (split muff)	none	low-medium	line shafting, easy removal
Flange	none (rigid)	high	turbines, heavy rigid drives
Bushed-pin	small angular & parallel	medium	motor - pump / fan
Oldham	parallel offset	low-medium	instrument, feed drives
Hooke's	angular ≤ 30°	medium-high	automobile propeller shaft
Gear / grid	1-3° angular	very high	rolling mills, heavy drives

- Catalogue selection : $T_{design} = K_s\, T_{rated}$; pick the smallest standard size with torque capacity ≥ T_{design} and bore range including d ; check max speed, misalignment and bore of both shafts.

6. Design of Flange Coupling

Two cast-iron (FG 200) flanges keyed to the shaft ends and held by n fitted bolts on a pitch circle. Standard proportions (IS 2949 / Bhandari) in terms of shaft dia d :



n = 3 (d ≤ 40) , 4 (40-100) , 6 (100-180)

Fig 2: Protected type flange coupling - standard proportions

Part	Dimension	Part	Dimension
Hub dia D_h	2 d	Bolt pitch circle D_1	3 d
Hub length L	1.5 d	Outer flange dia D_2	4 d
Flange thickness t	0.5 d	Protecting rim t_1	0.25 d
Key width w / thickness	d/4 ; 2w/3 (standard)	No. of bolts n	3, 4, 6

Design steps - check each part in shear (Torque T basis)

1. Shaft : $d^3 = 16\ T / \pi\ \tau_s$
2. Hub (hollow shaft in torsion) : $T = (\pi/16) \cdot (D_h^4 - d^4)/D_h \cdot \tau_h$
3. Key : shear $T = l\ w\ \tau_k\ d/2$, crushing $T = l\ (t/2)\ \sigma_c\ d/2$
4. Bolts (shear) : $T = n\ (\pi/4)\ d_b^2\ \tau_b \cdot D_1/2$
5. Flange at hub junction (shear) : $T = \pi\ D_h\ t\ \tau_f \cdot D_h/2$
6. Bolt crushing : $T = n\ d_b\ t\ \sigma_c \cdot D_1/2$

- Hub : with $D_h = 2d$: $T = 1.47\ d^3\ \tau_h$; flange : $T = 2\pi\ d^2\ t\ \tau_f$. Bolts are turned & fitted in reamed holes so that they are in shear, not tension (clearance bolts rely on friction - poor).

- Material : shaft & key steel (45C8 , 30C8) ; flange C.I. FG 200 ($\tau \approx 15$ MPa) ; bolts mild steel Fe 410. Typical allowable : τ_s = 40-60 , τ_{bolt} = 30 , τ_{key} = 40-50 , σ_c (key) = 80-100 , $\tau_{C.I.}$ = 15 MPa.

Example 3 : Flange coupling to transmit 15 kW at 600 rpm. $T = 9.55\times10^6 \times 15/600 = 238.75\times10^3$ N·mm. τ_s = 40 MPa.

Shaft : $d^3 = 16 \times 238750 / (\pi \times 40) = 30.4\times10^3 \rightarrow d = 31.2 \rightarrow$ 32 mm. Then $D_h = 64$, $L = 48$, $D_1 = 96$, $D_2 = 128$, $t = 16$, $t_1 = 8$ mm , $n = 3$.

Bolts : $238750 = 3 \times (\pi/4)\, d_b^2 \times 30 \times 48 \rightarrow d_b^2 = 70.4 \rightarrow d_b = 8.4$ mm → M10 fitted bolts. Key 10 × 8 × 48 : $\tau = 2T/(d\, l\, w) = 2\times238750/(32\times48\times10) = 31$ MPa < 40 (OK). Hub $\tau_h = 238750/(1.47 \times 32^3) = 5$ MPa (OK) ; flange $\tau_f = 238750/(2\pi \times 32^2 \times 16) = 2.3$ MPa (OK).

7. Muff (Sleeve) and Clamp Couplings - Rigid Types

Sleeve / muff coupling

A hollow cast-iron cylinder (muff) fits over the two shaft ends and is keyed to both ; simplest rigid coupling for light loads. Proportions : outer dia $D = 2d + 13$ mm ; length $L = 3.5\, d$; key along the full half-length ($L/2$ each shaft).

Muff (hollow shaft in torsion) : $T = (\pi/16) \cdot (D^4 - d^4)/D \cdot \tau_c$

Key (each shaft) : $\tau_k = 2T / (d\, (L/2)\, w) \le \tau_{allow}$

Split-muff / clamp (compression) coupling

Muff split into two halves clamped on the shafts by n bolts ; torque is transmitted by friction (a key is only for safety). No need to move shafts axially → easy mounting. Clamp bolts of total tension n W press the halves on the shaft :

$p = n\, W / (d\, L)$ (contact pressure on projected area)

$T = \mu\, p\, (\pi\, d\, L)\, (d/2) = (\pi/2)\, \mu\, n\, W\, d$

- $\mu \approx 0.25$ (steel on C.I.) ; bolt tension $W = \sigma_t\, A_t$ (core area) ; proportions : $D = 2d + 13$, $L = 3.5\, d$, n = 4-8 bolts. Use key as well if reversing torque or shock.

Example 7 : d = 40 mm, T = 400 N·m, μ = 0.25, 6 bolts : $W = 2T / (\pi\, \mu\, n\, d) = 2\times400\times10^3/(\pi \times 0.25 \times 6 \times 40) = 4244$ N per bolt → $A_t = 4244/60 = 71$ mm² (σ = 60 MPa) → M12 bolts.

8. Bushed-Pin Flexible Coupling

Modified flange coupling : the bolts are replaced by pins fitted with rubber bushes (leather / neoprene) on one flange ; the bush absorbs shock & allows ≈ 0.5-3° misalignment and axial float. Used for motor-pump, fan, compressor drives.

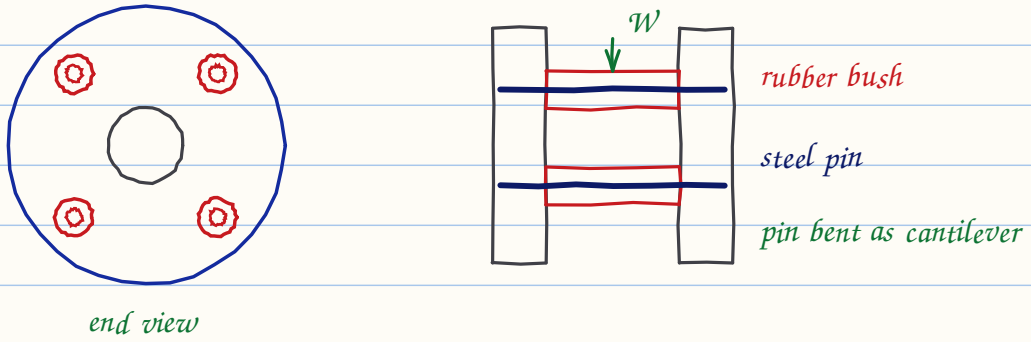


Fig 3: Bushed-pin flexible coupling

- Bush bearing pressure : $p_b = W / (d_2\, l) \le 0.5\ N/mm^2$ (rubber) ; W = load per pin = $2T / (n\, D_1) = 2T/(n\, D_p)$ where D_p = pitch circle dia of pins.
- Pin is treated as a cantilever loaded at the middle of the bush (length l) with a clearance $c \approx 5$ mm between flanges :

$$M = W\,(l/2 + c)$$

$$\sigma_b = 32\, M / \pi\, d_1^3 \le 25\text{-}35\ MPa$$

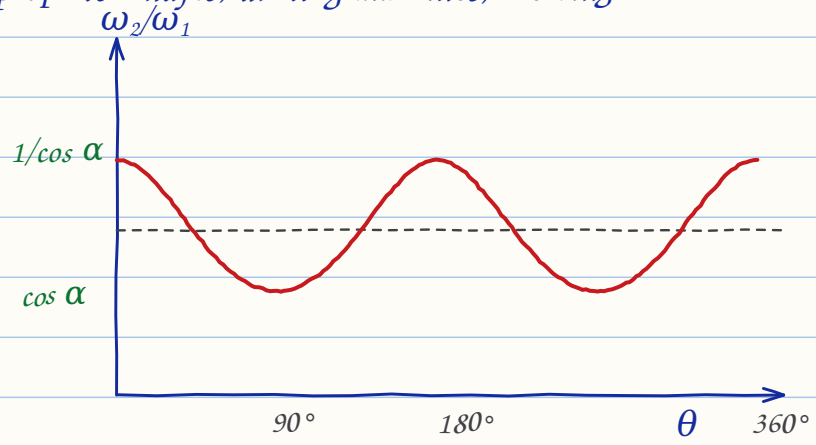
$$\tau = 4\, W / \pi\, d_1^2 \quad ; \quad \sigma_{eq} = \sqrt{(\sigma_b^2 + 3\tau^2)}$$

- Torque capacity : $T = W\, n\, D_p/2 = p_b\, d_2\, l\, n\, D_p/2$. Hub, key, shaft designed as in flange coupling. Number of pins 4-8 ; $d_2 = d_1 + 2$ (sleeve + rubber ≈ 6) ; pin is tapered on the driving flange side, nut & washer on the other.

Example 4 : Same $T = 238.75\times10^3$ N·mm, $d = 32$ mm. Take $n = 4$ pins on $D_p = 150$ mm : $W = 2T/(n\, D_p) = 2\times238750/(4\times150) = 796$ N. Bush $d_2 = 32$, $l = 50 \rightarrow p_b = 796/(32 \times 50) = 0.50\ N/mm^2$ (OK). $M = 796 \times (25 + 5) = 23.9\times10^3$ N·mm ; $\sigma_b = 30 \rightarrow d_1^3 = 32 \times 23900 / (\pi \times 30) = 8.1\times10^3 \rightarrow d_1 = 20$ mm ; (bush : $20 + 2\times6 = 32$ (OK)). $\tau = 4\times796/(\pi\, 20^2) = 2.5$ MPa - negligible.

9. Universal (Hooke's) Coupling

Joins two shafts whose axes intersect at an angle α (usually ≤ 20°, max 30-35°) : two forked yokes joined through a cross (spider) with four trunnions ; used in automobile propeller shafts, drilling machines, steering.



α = 30° : 0.866 ≤ ω_2/ω_1 ≤ 1.155

$\omega_2/\omega_1 = \cos\alpha / (1 - \sin^2\alpha \cos^2\theta)$

two cycles per revolution

Fig 4: Velocity ratio of a single Hooke's joint

- Even if shaft 1 runs at constant ω_1, driven shaft 2 has varying speed : $\omega_2 = \omega_1 \cos\alpha / (1 - \sin^2\alpha \cos^2\theta)$, θ = angle turned by driver from the position when the fork pins lie in the plane of shafts.
- Max $\omega_2 = \omega_1 / \cos\alpha$ (θ = 0°, 180°) ; min $\omega_2 = \omega_1 \cos\alpha$ (θ = 90°, 270°) ; speed fluctuation = $\omega_1 (1/\cos\alpha - \cos\alpha) = \omega_1 \sin\alpha \tan\alpha$. Equal at θ where $\tan\theta = \pm 1/\sqrt{(\cos\alpha)}$.
- Torque transmitted varies inversely : $T_2 = T_1 \omega_1/\omega_2$. Fluctuating speed → angular acceleration, torque ripple, vibration - gets worse as α increases.
- Double Hooke's joint (intermediate shaft) : angles at both joints equal and fork planes of the two joints in the same plane → the speed variation of the first joint is cancelled → $\omega_3 = \omega_1$ (constant velocity drive).

Design of the joint

Shaft : $d^3 = 16\,T / \pi\,\tau_s$

Pin (cross arm) load : $W = T / 2\,a$ (a = distance of trunnion from axis)

Pin : bearing $p = W / d_1\,l_1$; bending $M = W\,l_1/2 \rightarrow \sigma_b = 32\,M / \pi\,d_1^3$; shear $\tau = 4W$

- Yoke ears, boss & cross are checked in shear and tension ; use $T_{max} = T_1/\cos\alpha$ at the worst position ; pins are of 40C8 / case hardened, bearings needle roller or bushes.

Example 8 : Hooke's joint with $\alpha = 20°$, driving shaft at 300 rpm : $N_{2,max} = 300/\cos 20° = 319.2$ rpm ; $N_{2,min} = 300 \cos 20° = 281.9$ rpm ; variation = 37.3 rpm (≈ 12%). With a double joint (equal angles) → 300 rpm constant.

Example 9 : 10 kW at 300 rpm, $\tau_s = 40$ MPa : $T = 318\times10^3$ N·mm → $d^3 = 16\,T/\pi\,\tau = 40.5\times10^3$ → $d = 34.4$ → 35 mm. Trunnion arm at $a = 50$ mm → $W = T/2a = 3183$ N. Pin $d_1 = 18$ mm, $l_1 = 24$ mm : $p = W/d_1 l_1 = 7.4$ MPa (OK) ; $M = W\,l_1/2 = 38.2\times10^3$ → $\sigma_b = 32\,M/\pi\,d_1^3 = 67$ MPa (OK) ; $\tau = 4W/\pi\,d_1^2 = 12.5$ MPa (OK).

Other flexible couplings (short note)

- Oldham : two slotted discs + central disc with perpendicular tongues - parallel (offset) shafts ; speed constant, wear in slots.
- Gear (toothed) coupling : external teeth hubs in internal-tooth sleeve - high torque, allows 1-3° misalignment (steel mills).
- Chain : two sprockets enclosed by a double-strand chain - simple, cheap. Diaphragm / disc : metallic flexing discs - no lubrication, zero backlash.

Typical exam questions - Unit III

- (1) Derive the ASME code equation for a shaft under bending and torsion with shock factors ; state K_b, K_t values.
- (2) Design a shaft to transmit P kW at N rpm carrying a gear and pulley - draw B.M.D. in both planes.
- (3) Solid vs hollow shaft : compare strength, weight and torsional rigidity.
- (4) Design a protected-type flange coupling for given P, N ; sketch with dimensions.
- (5) Design a bushed-pin coupling ; why is rubber bush used ? What is checked in the pin ?
- (6) Hooke's joint : derive velocity ratio, explain why a double joint gives constant speed.
- (7) Differences between rigid and flexible coupling ; between clutch and coupling.

Item	Key formula
Torque	$T = 9.55\times10^6\ P / N$ (N·mm)
Shaft (ASME)	$d^3 = 16\sqrt{((K_bM)^2 + (K_tT)^2)}/(\pi\ \tau)$
Twist	$\theta° = 584\ T\ L / G\ d^4$
Flange bolts	$T = n\ (\pi/4)\ d_b^2\ \tau_b\ D_1/2$
Bush pressure	$p_b = W/(d_2\ l) \le 0.5$ MPa
Hooke's joint	$\omega_2/\omega_1 = \cos\alpha/(1 - \sin^2\alpha\ \cos^2\theta)$